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Evaluation

What is Evaluation?

After building the model, we must check:

- How accurate it is
- How reliable it is
- Whether it makes correct predictions

This stage is called Evaluation.

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Why Evaluation is Important?

Ensures model works correctly

✅Identifies mistakes

✅Improves model performance

✅Prevents wrong decisions

Without evaluation, we cannot trust the AI system.

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Understanding Predictions

- In many AI problems, the model predicts:

  - Yes / No

  - Positive / Negative

  - True / False

  - Pass / Fail

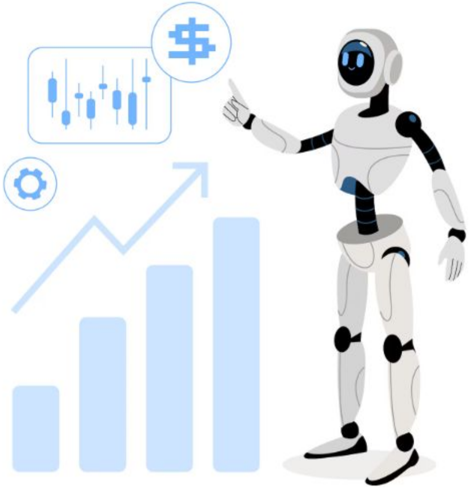
- To evaluate such models, we use 4 outcomes:

  1. True Positive

  2. False Positive

  3. True Negative

  4. False Negative



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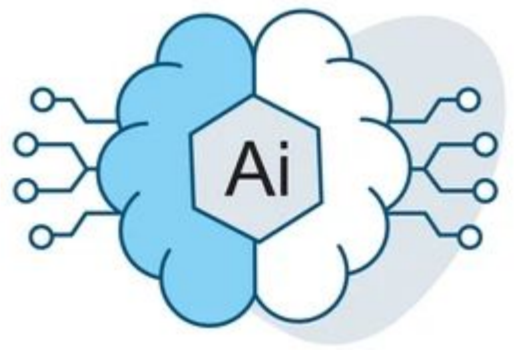
True Positive (TP)

True Positive means:

- The model predicted Positive, and the prediction was Correct.

Example:

- AI predicts a student will pass, and the student actually passes.
- Positive prediction + Correct result = True Positive



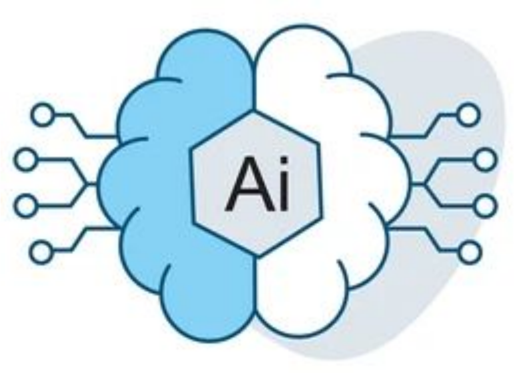
False Positive (FP)

False Positive means:

- The model predicted Positive, but the prediction was Wrong.

Example:

- AI predicts a student will pass, but the student actually fails.
- Positive prediction + Wrong result = False Positive



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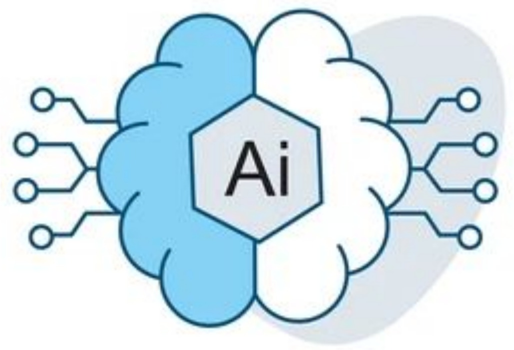
True Negative (TN)

True Negative means:

- The model predicted Negative, and the prediction was Correct.

Example:

- AI predicts a student will fail, and the student actually fails.
- Negative prediction + Correct result = True Negative



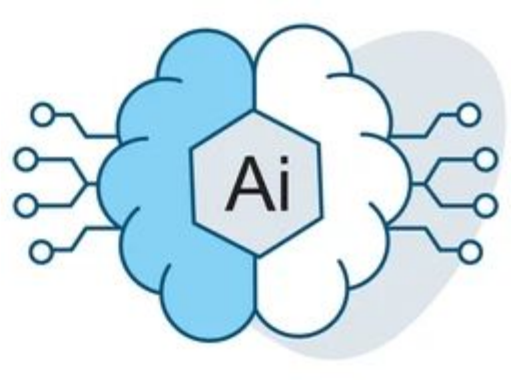
False Negative (FN)

False Negative means:

- The model predicted Negative, but the prediction was Wrong.

Example:

- AI predicts a student will fail, but the student actually passes.
- Negative prediction + Wrong result = False Negative



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Summary Table

Actual/Predicted	Positive	Negative
Positive	True Positive	False Negative
Negative	False Positive	True Negative

This table is called a Confusion Matrix.

Real-Life Example – Medical Test

Problem: Detecting Disease

✅ True Positive → Patient has disease & test says positive

✅ False Positive → Patient healthy but test says positive

✅ True Negative → Patient healthy & test says negative

✅ False Negative → Patient has disease but test says negative

✅ False Negative can be very dangerous in medical cases.

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Why These Terms Matter?

✅ Help measure accuracy

✅ Help reduce errors

✅ Help improve model performance

✅ Important in sensitive areas (healthcare, security)

Key Takeaways

✅ It checks how well the model performs.

✅There are 4 prediction outcomes:

- True Positive
- False Positive
- True Negative
- False Negative

✅ These are represented using a Confusion Matrix.

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THANK YOU

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